

PRADEEP VARMA PENUMATCHA

Senior Data Engineer

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linkedin.com/in/pradeep-penumatcha | HashiCorp Terraform Certified | 11+ Years Experience

PROFESSIONAL SUMMARY

Senior Data Engineer with 11+ years of experience building enterprise-grade batch and real-time data platforms across banking (JPMorgan Chase, Bank of America) and telecom domains. Deep expertise in PySpark, Spark Scala, AWS (EMR, Glue, S3, Athena), Databricks, Delta Lake, Apache Kafka, Airflow, and Terraform. Strong background in data lakehouse architecture, Spark performance optimization, and delivering ML-ready data pipelines at scale.

Notice Period: 60 (negotiable)

CORE TECHNICAL SKILLS

Distributed Processing: PySpark · Spark Scala · Spark SQL · Spark Streaming · Hadoop · HDFS · YARN

Cloud & Lakehouse: AWS (EMR, Glue, S3, Athena, Lambda) · Databricks · Delta Lake · Medallion Architecture

Streaming & Orchestration: Apache Kafka · Spark Structured Streaming · Apache Airflow · Control-M · AutoSys

Languages & Databases: Python · Scala · SQL · PostgreSQL · MySQL · Oracle · Cassandra · Apache Hive

DevOps & IaC: Terraform (HashiCorp Certified) · CI/CD · Git · GitHub Actions

Practices: ETL/ELT · Data Modeling (Star Schema, SCD) · Data Quality · Feature Engineering · Agile/Scrum

PROFESSIONAL EXPERIENCE

Senior Data Engineer (SSE III) | JPMorgan Chase & Co. *Hyderabad · Sep 2021 – Present*

- Architected PySpark ETL/ELT pipelines on AWS EMR processing multi-TB daily datasets for analytics and ML feature stores.
- Tuned Spark jobs via partitioning, broadcast joins, and shuffle reduction — cutting runtimes and AWS compute cost significantly.
- Built Databricks workloads with Delta Lake medallion (bronze/silver/gold) architecture for versioned, auditable data layers.
- Provisioned AWS infrastructure using Terraform (certified), enabling repeatable IaC deployments across dev/staging/prod environments.
- Orchestrated workflows with Apache Airflow and Control-M; integrated S3, RDBMS, and REST APIs with data quality validation.
- Delivered feature engineering pipelines and ML-ready datasets, partnering with Data Scientists for model training and inference.

Data Engineer (Apps Analyst II) | BA Continuum – Bank of America *Hyderabad · Jul 2019 – Sep 2021*

- Migrated Oracle/SQL Server data to Hadoop/Spark via Apache Sqoop for banking regulatory analytics at scale.
- Built Spark Scala curated datasets; designed dimensional models (star schema, SCD Type 1 & 2) in Hive and Spark SQL.
- Integrated Apache Kafka for near-real-time ingestion; tuned Hive/Spark SQL queries for reporting performance improvements.

Senior Software Engineer | Tech Mahindra Pvt Ltd *Hyderabad · Aug 2018 – Jul 2019*

- Built Kafka + Spark Streaming real-time pipelines for telecom network analytics; stored events in Apache Cassandra for low-latency dashboards.
- Delivered sub-second latency insights for network operations teams supporting SLA monitoring and fault detection.

Big Data Engineer (Analyst) | Cognizant Technology Solutions *Hyderabad · May 2014 – Jul 2018*

- Developed batch pipelines (Sqoop, Hive, Spark Scala), Kafka + Flume real-time ingestion; data stored in ORC/Parquet for BI and reporting.

KEY ACHIEVEMENTS

- Reduced ETL pipeline runtime through Spark partitioning and shuffle optimization for multi-TB daily batch jobs at JPMorgan Chase.
- Architected Delta Lake medallion data lakehouse on Databricks, enabling reliable, versioned data for ML workloads.
- Automated AWS infrastructure with Terraform eliminating manual configuration drift across environments.
- Migrated enterprise RDBMS to Hadoop/Spark at Bank of America, enabling scalable analytics and faster regulatory reporting.

CERTIFICATIONS

- HashiCorp Certified: Terraform Associate — IaC for cloud-agnostic deployments (AWS/Azure/GCP)
- Databricks Hands-On — Delta Lake, Spark Tuning, Databricks Jobs, Unity Catalog, Notebooks

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science / Engineering

Andhra Pradesh, India

TECHNICAL KEYWORDS

Apache Spark · PySpark · Spark Scala · AWS EMR · AWS Glue · S3 · Athena · Databricks · Delta Lake · Kafka · Airflow · Terraform · Hadoop · Hive · Python · SQL · ETL · ELT · Data Lakehouse · Data Pipeline · Data Modeling · Cassandra · PostgreSQL · Oracle · CI/CD · Git · Real-Time Streaming · Batch Processing · Feature Engineering · Big Data · Banking · Telecom · ML Pipelines